

HR Datamart Entity-Relationship Overview

Warehouse: PostgreSQL per tenant (hrm_wh_{tenant_id})

Architecture: Medallion (hr_raw -> hr -> hr_semantic) + Kimball star schema

Metadata: docs\datamart-er.yml

Domains: Enterprise HR, Leave Intelligence (Phases 0-4), Payroll Phase 6

Legend

DIM - Dimension (conformed, surrogate keys)

FACT - Atomic grain fact table

MART - BI-ready aggregate / denormalized report table

VIEW - Semantic or legacy compatibility layer

Medallion data flow

Bronze (hr_raw): ETL staging from MintHRM source (stg_*)

Silver/Gold (hr): dbt dimensions, facts, and marts (*_sk surrogate keys)

Semantic (hr_semantic): Stable vw_* views for APIs and reports

Control (hr_control): ETL log, watermarks, data_dictionary_* (auto-sync after dbt)

Conformed dim_employee and dim_date are shared across Enterprise, Leave, and Payroll.

1. Enterprise HR - Entities (schema: hr)

Object	Kind	Primary key	Grain / role
dim_date	DIM	date_sk	Calendar
dim_designation	DIM	designation_sk	Job title / grade
dim_employee	DIM	employee_sk	SCD2 employee version (conformed); superior_emp_no = immediate manager emp_no
dim_org_unit	DIM	org_unit_sk	Organization hierarchy
dim_shift	DIM	shift_sk	Work shift
fct_daily_attendance	FACT	daily_attendance_sk	
fct_employment_snapshot	FACT	employment_snapshot_sk	
fct_lifecycle_event	FACT	lifecycle_event_sk	
fct_overtime	FACT	overtime_sk	
fct_salary_change	FACT	salary_change_sk	
mart_attendance_monthly_summary	MART	-	Employee x month attendance
mart_employee_current	MART	-	Current active employee OBT; resolves supervisor via superior_emp_no -> dim_employee.emp_no
mart_headcount_monthly	MART	-	Org x month headcount
mart_salary_band_summary	MART	-	Band x period summary

Enterprise HR - Relationships

From (fact/mart)	To (dim/upstream)	Join key
fct_daily_attendance	dim_employee	employee_sk
fct_overtime	dim_employee	employee_sk
fct_employment_snapshot	dim_employee	employee_sk
fct_employment_snapshot	dim_designation	designation_sk
fct_employment_snapshot	dim_org_unit	org_unit_sk
fct_lifecycle_event	dim_employee	employee_sk
fct_salary_change	dim_employee	employee_sk
mart_employee_current	dim_employee	employee_sk
mart_employee_current	dim_employee	superior_emp_no -> emp_no
mart_headcount_monthly	fct_employment_snapshot	aggregates
mart_headcount_monthly	dim_date	date_sk
mart_attendance_monthly_summary	fct_daily_attendance	aggregates
mart_attendance_monthly_summary	dim_employee	employee_sk
mart_salary_band_summary	dim_employee	employee_sk

2. Leave Intelligence Mart - Entities (schema: hr)

Phases 0-4 implemented. Legacy fact_leave_balance bridges to fct_leave_application.

Object	Kind	Primary key	Grain / role
dim_date	DIM	date_sk	Calendar
dim_employee	DIM	employee_sk	SCD2 employee version (conformed); superior_emp_no = immediate manager emp_no
dim_leave_approval_level	DIM	leave_approval_level_sk	Approval hierarchy lookup
dim_leave_reason	DIM	leave_reason_sk	Predefined leave purpose lookup
dim_leave_source	DIM	leave_source_sk	Leave module origin lookup
dim_leave_status	DIM	leave_status_sk	Normalized workflow status lookup
dim_leave_type	DIM	leave_type_sk	SCD2 leave type (hr_leavetype)
fact_leave_balance	FACT	id	Legacy approved-leave summary over fct_leave_application
fct_leave_application	FACT	leave_application_sk	One row per leave application (all implemented modules)
fct_leave_approval	FACT	leave_approval_sk	One row per approval action
fct_leave_balance_snapshot	FACT	leave_balance_snapshot_sk	Employee x leave type x snapshot date
fct_leave_daily	FACT	leave_daily_sk	Employee x leave date
fct_leave_planner	FACT	leave_planner_sk	One row per planned leave
fct_lieu_leave_entitlement	FACT	lieu_leave_entitlement_sk	Earned lieu entitlement event
fct_maternity_leave	FACT	maternity_leave_sk	One row per maternity leave request
fct_short_leave_usage	FACT	short_leave_usage_sk	One row per short leave request
mart_department_leave_heatmap	MART	-	Department x calendar date occupancy
mart_leave_employee_profile	MART	-	Employee leave behavior profile
mart_leave_monthly_summary	MART	-	Month x department x leave type

Leave Intelligence - Relationships

From (fact/mart)	To (dim/upstream)	Join key
fact_leave_balance	dim_employee	employee_sk
fact_leave_balance	fct_leave_application	legacy over
fct_leave_application	dim_employee	employee_sk
fct_leave_application	dim_leave_type	leave_type_sk
fct_leave_application	dim_leave_status	leave_status_sk
fct_leave_application	dim_leave_source	leave_source_sk
fct_leave_application	dim_leave_reason	leave_reason_sk
fct_leave_application	dim_date	start_date_sk
fct_leave_daily	fct_leave_application	leave_application_sk
fct_leave_daily	dim_employee	employee_sk
fct_leave_daily	dim_date	date_sk
fct_leave_balance_snapshot	dim_employee	employee_sk

From (fact/mart)	To (dim/upstream)	Join key
fct_leave_balance_snapshot	dim_leave_type	leave_type_sk
fct_leave_balance_snapshot	dim_date	snapshot_date_sk
fct_leave_approval	fct_leave_application	leave_application_sk
fct_leave_approval	dim_employee	approver_employee_sk
fct_leave_approval	dim_leave_approval_level	leave_approval_level_sk
fct_short_leave_usage	dim_employee	employee_sk
fct_short_leave_usage	dim_leave_type	leave_type_sk
fct_lieu_leave_entitlement	dim_employee	employee_sk
fct_maternity_leave	dim_employee	employee_sk
fct_maternity_leave	dim_leave_type	leave_type_sk
fct_leave_planner	dim_employee	employee_sk
fct_leave_planner	fct_leave_application	actual_leave_application_sk
mart_leave_monthly_summary	fct_leave_daily	aggregates
mart_leave_employee_profile	fct_leave_application	aggregates
mart_department_leave_heatmap	fct_leave_daily	aggregates

3. Payroll Phase 6 - Entities (schema: hr)

Object	Kind	Primary key	Grain / role
dim_bank	DIM	bank_sk	Bank master
dim_bank_branch	DIM	bank_branch_sk	Bank branch master
dim_canonical_pay_item	DIM	canonical_pay_item_sk	Normalized pay components
dim_employee	DIM	employee_sk	SCD2 employee version (conformed); superior_emp_no = immediate manager emp_no
dim_payroll_group	DIM	payroll_group_sk	Payroll run group
dim_payroll_period	DIM	payroll_period_sk	Year x month x half
fact_payroll	VIEW	-	Legacy view of fct_processed_salary
fct_compliance_payroll	FACT	compliance_payroll_sk	Statutory slice
fct_noncash_benefit	FACT	noncash_benefit_sk	Non-cash benefit
fct_payroll_process_execution	FACT	payroll_process_execution_sk	Run metadata
fct_processed_add_ded	FACT	processed_add_ded_sk	Add/ded line
fct_processed_attendance_payroll	FACT	processed_attendance_payroll_sk	Attendance in payroll
fct_processed_loan_deduction	FACT	processed_loan_sk	Loan deduction
fct_processed_multi_currency_salary	FACT	processed_multi_currency_salary_sk	FX salary
fct_processed_salary	FACT	processed_salary_sk	Emp x period x group
fct_processed_tax	FACT	processed_tax_sk	Tax component
fct_salary_bank_instruction	FACT	salary_bank_instruction_sk	Salary bank payment instruction by employee x period x account
fct_variable_pay_item	FACT	variable_pay_item_sk	Variable pay line
mart_cost_to_company	MART	-	Cost to company (CTC)
mart_dynamic_pay_items	MART	-	Dynamic pay items
mart_horizontal_paysheet_core	MART	-	Wide paysheet core
mart_horizontal_paysheet_dynamic	MART	-	Pivoted pay items
mart_payroll_pivot_column_catalog	MART	-	Tenant x pivot_key dynamic paysheet column catalog
mart_payroll_reconciliation	MART	-	Group x period reconcile
mart_processed_payroll_summary	MART	-	Emp x period BI summary
mart_statutory_summary	MART	-	Statutory rolup

Payroll Phase 6 - Relationships

From	To	Join key / role
fct_processed_salary	dim_employee	employee_sk
fct_processed_salary	dim_payroll_period	payroll_period_sk
fct_processed_salary	dim_payroll_group	payroll_group_sk
fct_processed_add_ded	dim_employee	employee_sk
fct_processed_add_ded	dim_payroll_period	payroll_period_sk
fct_processed_add_ded	dim_payroll_group	payroll_group_sk
fct_processed_add_ded	dim_canonical_pay_item	canonical_pay_item_sk

From	To	Join key / role
fct_variable_pay_item	dim_employee	employee_sk
fct_variable_pay_item	dim_payroll_period	payroll_period_sk
fct_variable_pay_item	dim_canonical_pay_item	canonical_pay_item_sk
fct_processed_tax	dim_employee	employee_sk
fct_processed_tax	dim_payroll_period	payroll_period_sk
fct_processed_loan_deduction	dim_employee	employee_sk
fct_processed_loan_deduction	dim_payroll_period	payroll_period_sk
fct_compliance_payroll	dim_employee	employee_sk
fct_compliance_payroll	dim_payroll_period	payroll_period_sk
fct_salary_bank_instruction	dim_employee	employee_sk
fct_salary_bank_instruction	dim_payroll_period	payroll_period_sk
fct_salary_bank_instruction	dim_payroll_group	payroll_group_sk
fct_salary_bank_instruction	dim_bank	bank_sk
fct_salary_bank_instruction	dim_bank_branch	bank_branch_sk
fct_payroll_process_execution	dim_payroll_period	payroll_period_sk
fct_payroll_process_execution	dim_payroll_group	payroll_group_sk
mart_processed_payroll_summary	fct_processed_salary	rolls up
mart_horizontal_paysheet_core	fct_processed_salary	denormalizes
mart_horizontal_paysheet_dynamic	mart_horizontal_paysheet_core	extends
mart_horizontal_paysheet_dynamic	fct_processed_add_ded	pivot
mart_horizontal_paysheet_dynamic	fct_variable_pay_item	pivot
mart_horizontal_paysheet_dynamic	mart_payroll_pivot_column_catalog	column catalog
mart_payroll_reconciliation	fct_processed_salary	reconciles
mart_statutory_summary	fct_compliance_payroll	aggregates
mart_cost_to_company	fct_processed_salary	extends
mart_cost_to_company	fct_noncash_benefit	extends
mart_dynamic_pay_items	fct_variable_pay_item	aggregates
mart_dynamic_pay_items	fct_processed_add_ded	aggregates
fact_payroll	fct_processed_salary	view of

4. Semantic layer (schema: hr_semantic)

Consumer-facing dbt views. Physical marts may change; views provide a stable API contract.

View	Upstream marts / facts
vw_payroll_summary	mart_processed_payroll_summary, payroll facts
vw_headcount	mart_headcount_monthly, dim_employee
vw_employment_monthly	fct_employment_snapshot
vw_attendance_summary	mart_attendance_monthly_summary
vw_salary_bands	mart_salary_band_summary
vw_lifecycle_summary	fct_lifecycle_event
vw_turnover	lifecycle / employment facts
vw_leave_summary	fact_leave_balance, fct_leave_application, dim_employee
vw_current_leave_balance	fct_leave_balance_snapshot, dim_employee, dim_leave_type
vw_employee_leave_history	fct_leave_application, dim_employee, dim_leave_type
vw_pending_leave_approvals	fct_leave_approval, fct_leave_application, dim_employee
vw_leave_liability	fct_leave_balance_snapshot, dim_employee, dim_leave_type
vw_leave_utilization	mart_leave_monthly_summary, fct_leave_balance_snapshot
vw_performance_summary	performance staging

5. Control plane - Data dictionary (schema: hr_control)

ETL-maintained metadata tables. Refreshed after each successful dbt run (SYNC_DATA_DICTIONARY_ON_ETL).

Object	Kind	Primary key	Grain / role
data_dictionary_object	TABLE	object_id	One row per warehouse table/view per tenant
data_dictionary_field	TABLE	field_id	One row per column per object

Data dictionary - Semantic exposure

View	Reads
vw_data_dictionary	data_dictionary_object JOIN data_dictionary_field (flat field catalog)